

Development and technical directions of new power systems

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Committee of the North China Electric Power
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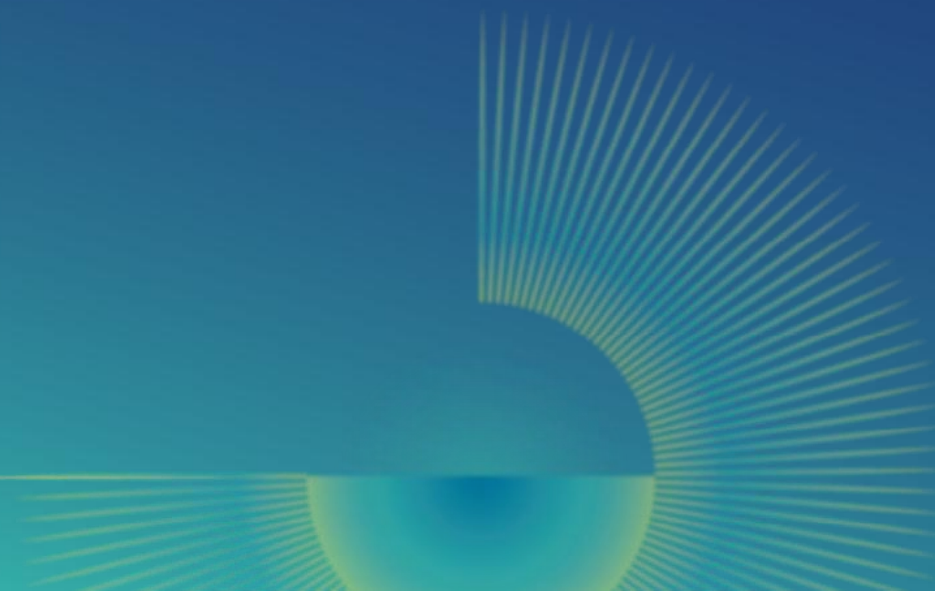
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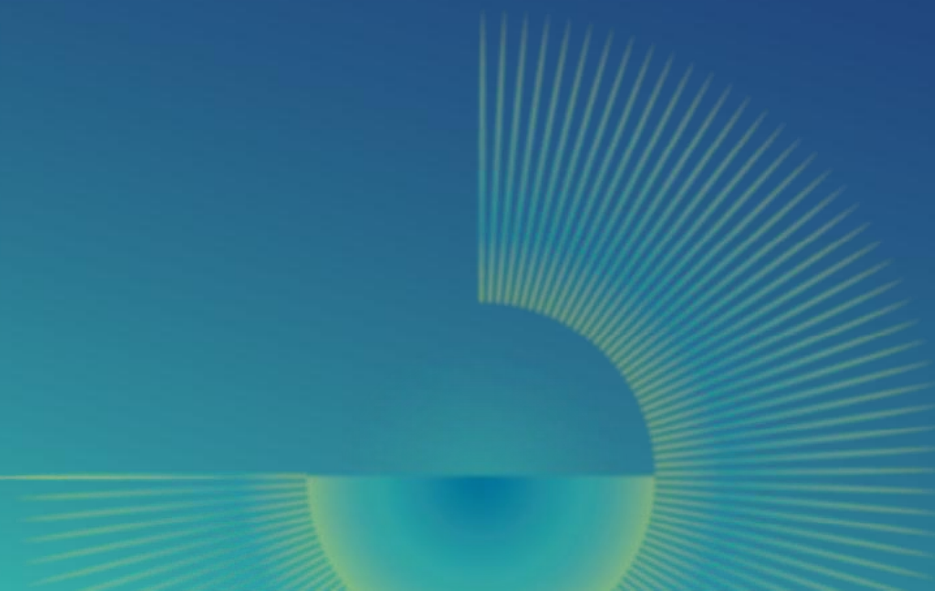
Development and technical directions of new power systems

新型电力系统发展与技术方向

1. Main features of the new power system
新型电力系统的主要特征
2. Technical directions for new power systems
新型电力系统的技术发展方向
3. Key application technologies for new power systems
新型电力系统的关键应用技术

ONE

Main features of the new power system



On March 15, 2021, the Central Financial and Economic Commission of CPC was held to study the basic ideas and key measures for achieving carbon peaking and carbon neutrality.

2021年3月15日，中央财经委员会，研究实现碳达峰、碳中和的基本思路和主要举措

Carbon peaking before 2030, carbon neutrality before 2060

2030年前碳达峰、2060年前碳中和

Build a new power system dominated by new energy (with the proportion of new energy increasing year by year).

深化电力体制改革，构建以新能源为主体（新能源占比逐年提高的）的新型电力系统

Basic characteristics of energy in China

能源的基本特征

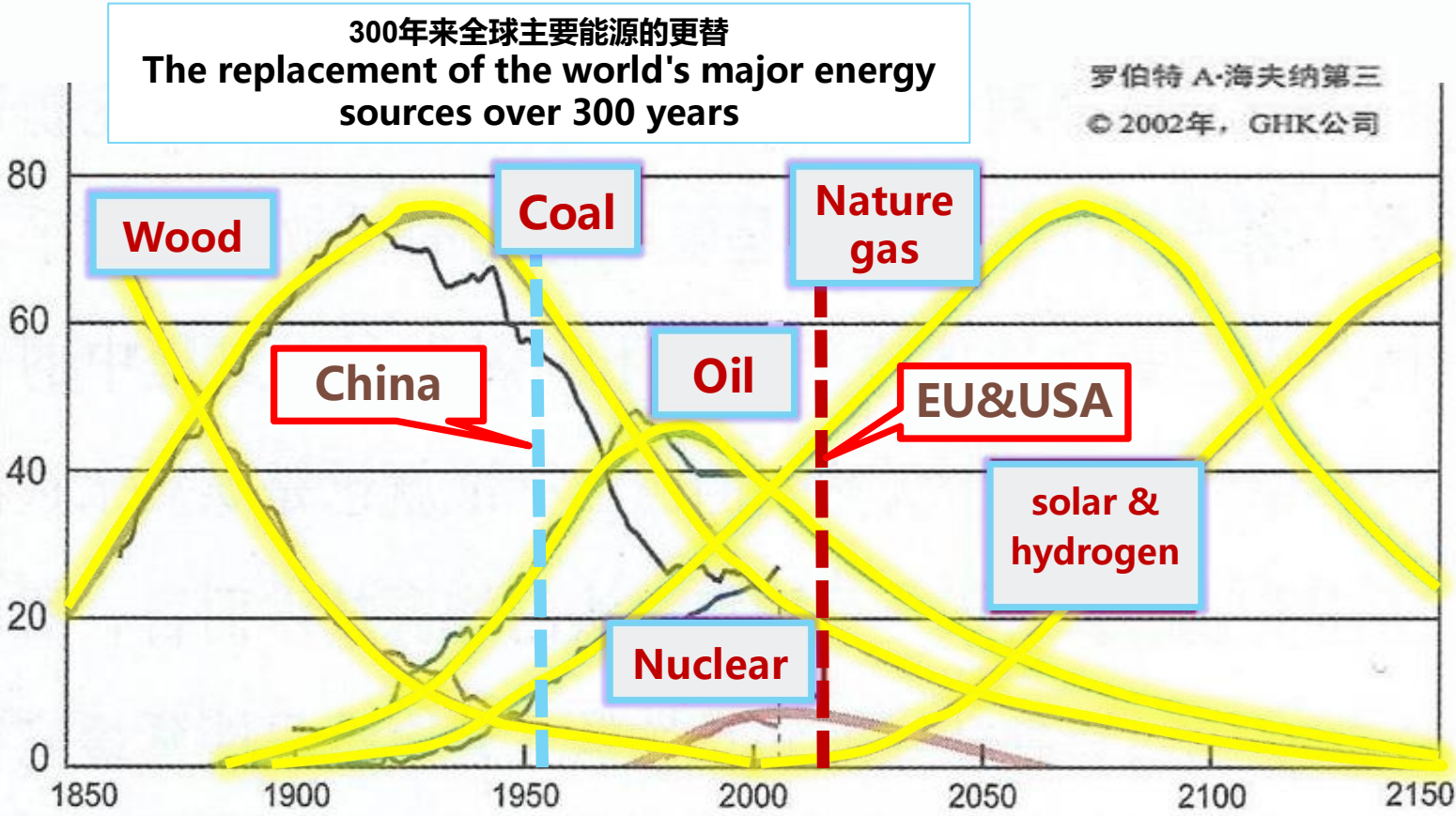
- The coal-dominated energy structure **will not change significantly in the short term**
- The per capita share of energy resources **is low in China**
- Energy is distributed in reverse direction, **energy resources is far away from the load center**

能源系统的基本特征

1. 以煤为主的能源结构短期内难以改变
2. 能源资源的人均占有率低
3. 能源逆向分布，远离负荷中心

Energy Structure is Based on Coal

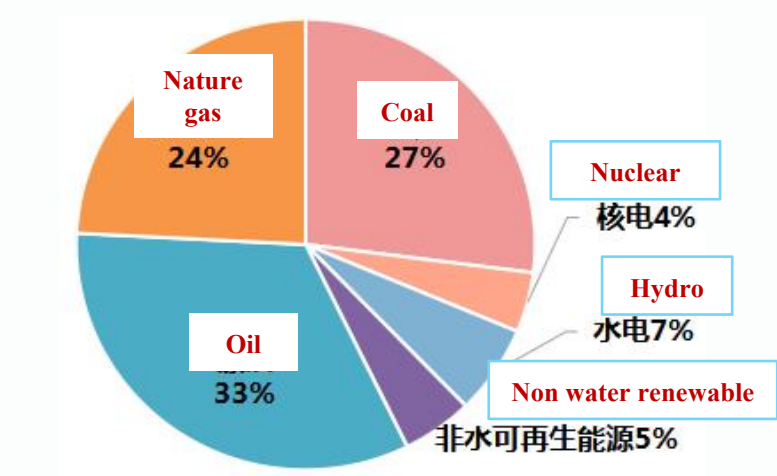
能源特征-1：能源结构以煤为主



大力发展非化石能源，改变以煤为主的能源结构

Vigorously develop non-fossil energy sources and change the situation that energy structure is dominated by coal

全球一次能源消费 (2019年)
Global primary energy consumption (2019)



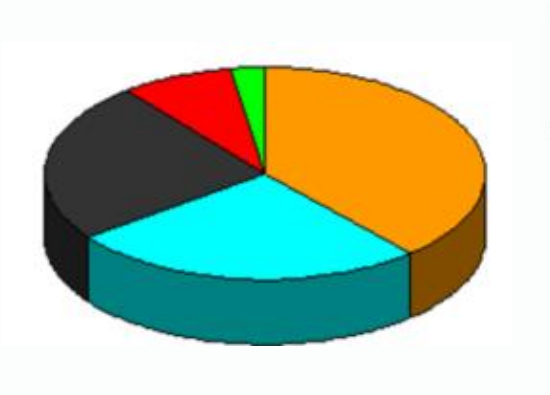
中国一次能源产销量 (2019)
Production and consumption of Chinese primary energy (2019)

	Coal (10 ⁸ tons)	Oil (10 ⁸ tons)	Nature gas (10 ⁸ m ³)
production	27.23	1.91	1761
consumption	27.87	6.96	3073
occupy	57.70%	18.90%	7.8%
import	0.64	5.05	1312

China's energy characteristics -1

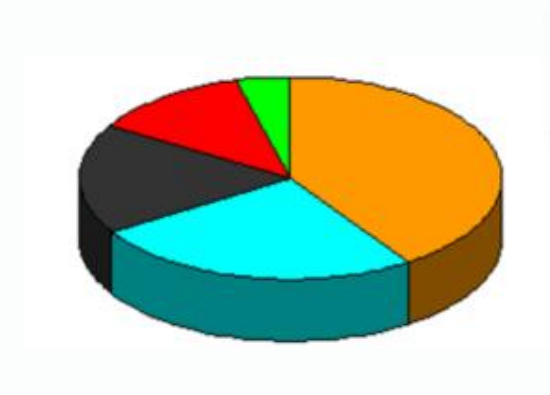
Energy Structure is Based on Coal

能源特征-1：能源结构以煤为主



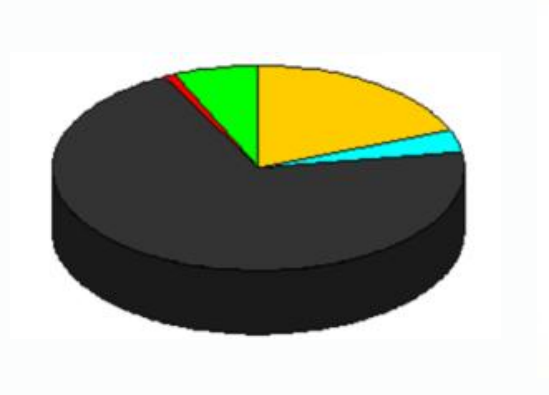
Oil	884.5	38.48%
Nature Gas	600.7	26.13%
Coal	565	24.58%
Nuclear	192	8.35%
Hydor	56.7	2.47%
Total	2,299	100%

USA



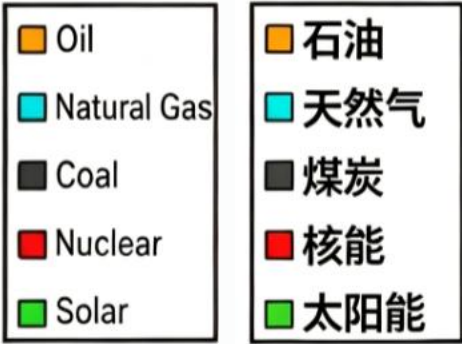
Oil	702.6	40.66%
Nature Gas	441.1	25.52%
Coal	301.2	17.43%
Nuclear	212.7	12.31%
Hydor	70.6	4.08%
Total	1,728.20	100%

EU



Oil	375.71	18.90%
Nature Gas	72.63	8.10%
Coal	1406.29	57.70%
Nuclear	15.48	1.70%
Hydor	132.41	13.60%
Total	2,002.52	100%

China

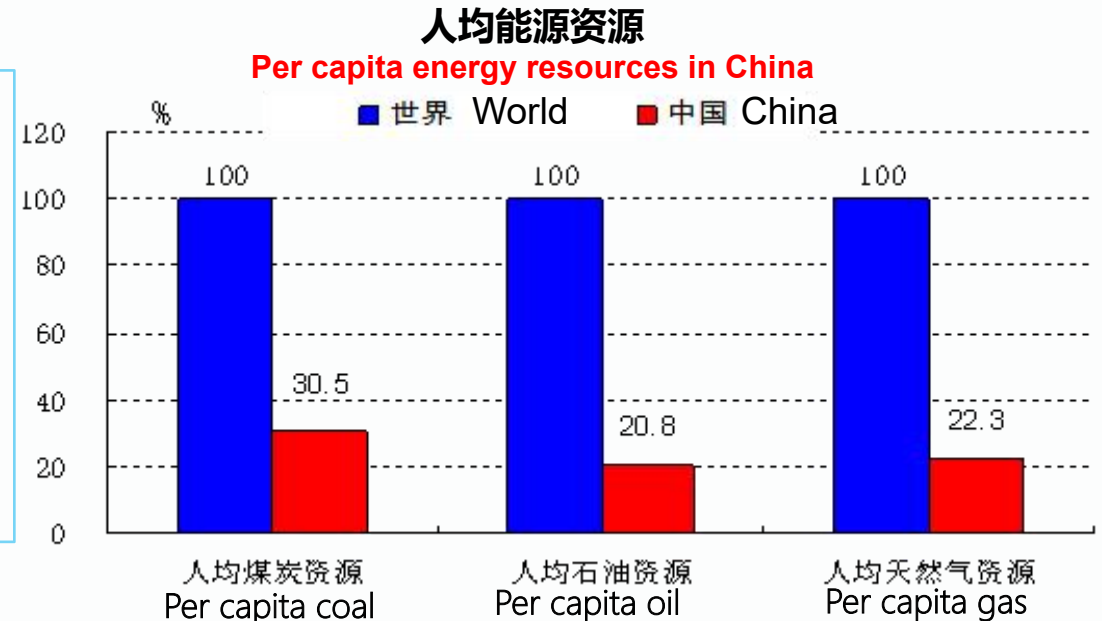


China's energy characteristics - 2: Average Energy Resources Per Capita is Low

能源特征-2: 能源资源的占有率低

Although China's total energy resource is rich, hydro power ranks the first and coal ranks the second, but the per capita energy is poor.

- The per capita coal is 30% of the world average
- The per capita oil is 20% of the world average
- The per capita natural gas is 22% of the world average



China's energy characteristics-3...The energy distribution is far from the load center

能源分布远离负荷中心

- ◆ **The electric power flow is primarily transmitted from the northwestern coal-fired power, centralized renewable energy bases, and from the southwestern hydropower base, to the eastern regions, by long transmission distances and large transmission capacities.**
- ◆ **Energy flow of natural gas is also from west to east: natural gas from Xinjiang, Russia and Central Asia is transported to the receiving center of Central and Eastern China by the West-to-East Gas Pipeline.**
- ◆ **电能量流主要由西北煤电和集中式可再生能源基地、西南水电基地送往中东部，输送距离长、输送容量大。**
- ◆ **天然气能量流：由西气东输管道将新疆、俄罗斯、中亚等地的天然气送往中东部受端中心。**

On generation side, clean substitution is implemented to create a zero-carbon powersystem 电力生产实施清洁替代，打造零碳电力系统

◆ We have to make clean substitution on the production side to build a deep zero-carbon power system.

The thermal coal power plant consumes about 1.7 bilion tons, with 4.5 billion tons of CO₂ emissions. The fundamental way to reduce carbon emissions lies in the steady replacement of coal power generation by clean energy, and build a power supply system based on renewable energy, nuclear power and biomass power generation, etc.

发电用煤占煤炭消耗的一半、约17亿吨标煤，产生CO₂ 排放45亿吨。减碳的根本途径在于清洁能源对煤炭发电的稳步替代，形成以新能源、水电、核电、生物质发电等为主的电力供应体系。

◆ By 2060, thermal coal consumption will reduce to about 300 million tons, and the remaining percentage of CO₂ emissions from fossil energy generation will be neutralized through carbon reduction technologies, such as carbon sink and CCUS.

2060年，电煤消耗降至3亿吨左右，产生的CO₂排放通过碳汇和CCUS 等技术中和。

Power system is key platform to achieve carbon neutrality goals

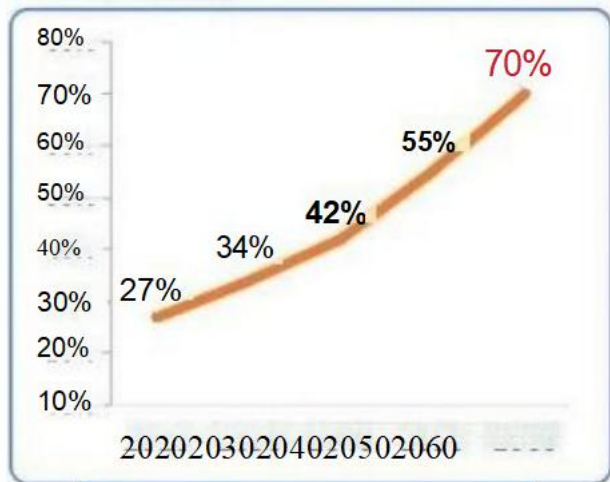
电力系统是实现碳中和的枢纽平台

- ◆ To achieve the "dual carbon" goal, it is necessary to reduce fossil energy consumption and increase non-fossil energy consumption. Non-fossil energy is mainly converted into electricity use, and power system will play an important role of the conversion.

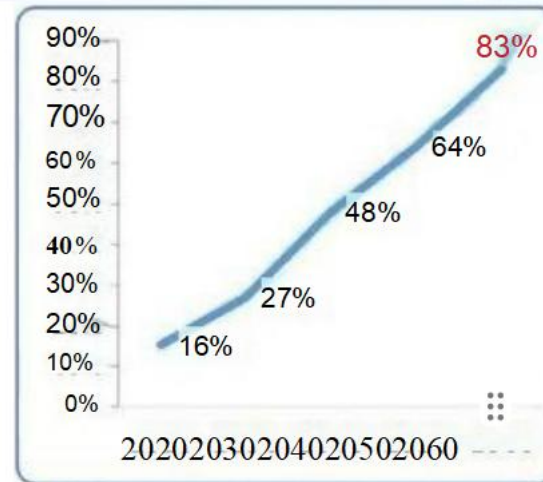
实现“双碳”目标，根本上要减少化石能源消费、大幅增加非化石能源消费。非化石能源主要转化为电力使用，电力系统将发挥转换枢纽作用。

- ◆ By 2060, China's energy transition will have achieved its "70/80/90" targets, in which the renewable energy accounts for more than 60% of power generation.

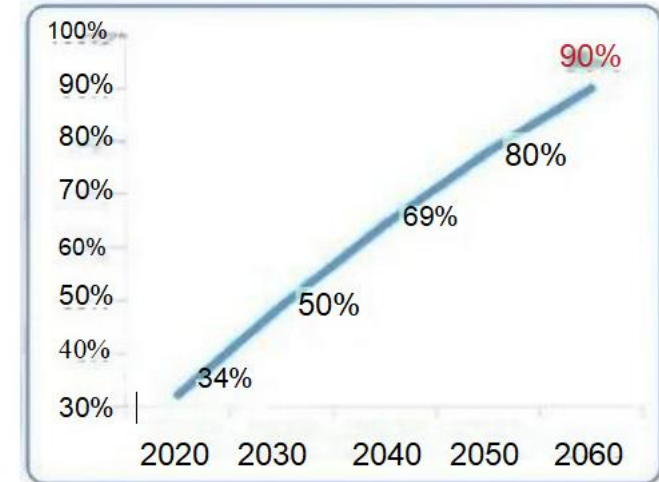
到2060年，中国能源电力转型将实现“70/80/90”目标。其中新能源发电量占比超过60%。



Proportion of electricity energy consumption
电能消费比重



Proportion of non-fossil energy consumption
非化石能源消费比重



Proportion of clean energy power generation
清洁能源发电比重

New power system with renewable energy as the main body

新能源为主体的新型电力系统

第一代（19世纪末-20世纪60年代）

- ◆ 初级发展模式，电源和电网初步建设



第二代（20世纪70年代-21世纪初）

- ◆ 高度依赖化石能源，不可持续发展



第三代（21世纪初-21世纪中叶）

- ◆ 非化石能源为主的能源电力系统

The first generation of power system (1900-1970s)

Primary development mode, preliminary construction of power supply and power grid



The second generation of power system (1970s-2000)

Highly dependent on fossil energy and unsustainable development



The third generation of power system (2000-2050)

Non fossil energy-based power system -- New power system with renewable energy as the main body

	First generation	Second generation	Third generation
Main features	● Small units (coal, oil and gas, small hydropower) ● Low voltage (220kV and below) ● Small power grid (urban power grid, isolated power grid)	● Large units (coal, hydropower, nuclear power) ● High voltage (750kV and below) ● Large power grid (AC / DC hybrid power grid)	● Power generation based on renewable energy and clean energy (more than 60% - 70%) ● Combination of large scale power plants and distributed power generation ● Combination of backbone power grid, local distribution network and Microgrid
Control protection	Simple protection and manual control	● Fast protection and optimal control ● Fast removal of faults in power transmission and transformation equipment	● Smart grid control and protection system ● Intelligent self-healing of power transmission and transformation equipment and network
Transmission mode	Overhead line and cable transmission below 220kV	● Ultra high voltage AC / DC transmission ● Overhead power transmission mode is mainly used	● Large capacity, low loss and environment-friendly transmission mode
Dispatching and energy management	Empirical scheduling	● Analytical scheduling ● Power supply side energy management system adapting to load change	● random variation of adaptive source load ● Centralized- decentralized intelligent scheduling ● Integrated energy management system with full consideration of demand side response, energy storage and electric vehicle charge and discharge
Safety and reliability	Low power grid security and power supply reliability	● The safety and reliability are improved, and the risk of power failure in large power grid still exists	● The reliability of power supply has been greatly improved, and the risk of accidental power failure of users has been basically eliminated
Resource allocation mode	Poor ability to optimize the allocation of economic resources	● Large unit and large power grid economies of scale ● Ability to optimize the allocation of large-scale resources ● Electricity market	● Combination of concentration and distribution ● Economies of scale ● Integrated energy and power services ● Integrated energy power market
Distribution network service	One way power flow, single power service	● Unidirectional power flow in distribution network ● Single power service	● Intelligent active distribution network and Microgrid with wide access to distributed energy, two-way energy and power flow ● Energy, power and information services
Development model	Development model in the primary stage	● Highly dependent on fossil energy ● Unsustainable development model	● Based on renewable and clean energy ● Sustainable energy and power development mode

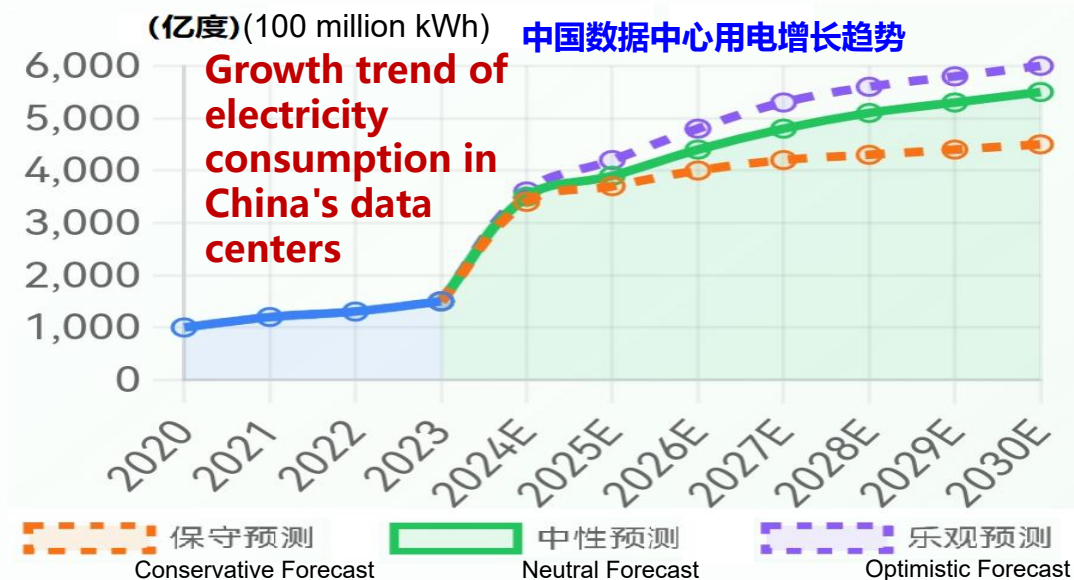
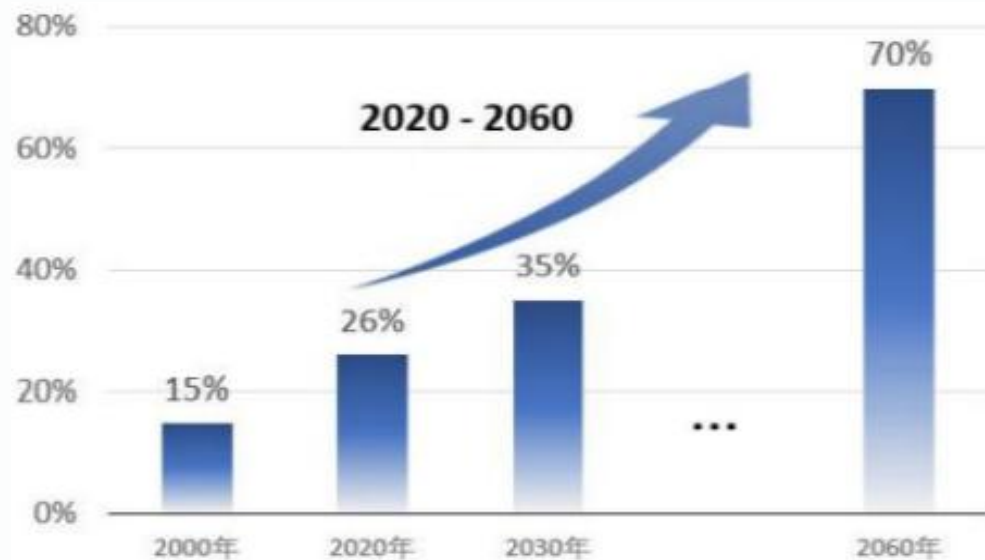
New Power System Feature 1: Electricity demand continues to grow

新型电力系统的主要特征-1 电力电量需求仍持续增长

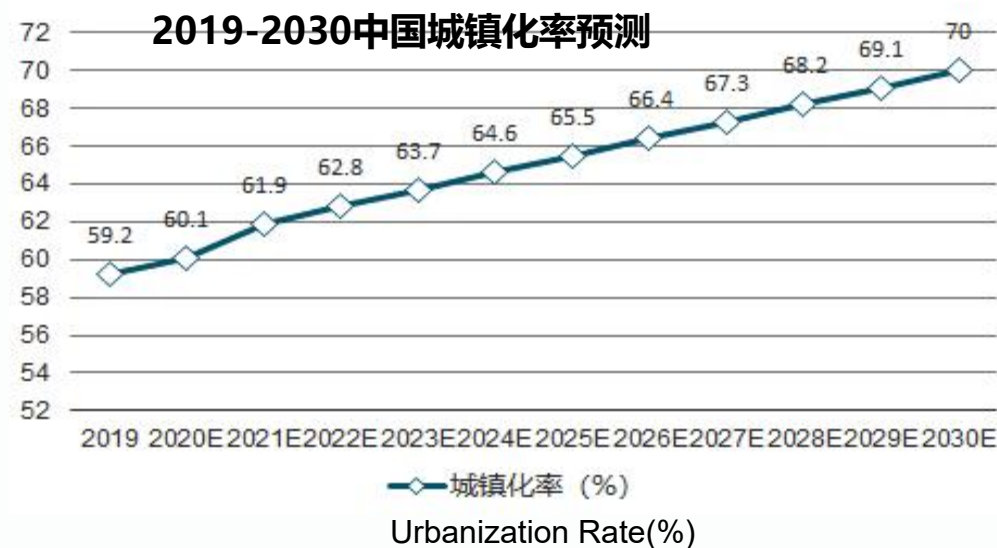
In 2015, China's per capita electricity consumption was 4000 kWh

In 2025, China's per capita electricity consumption is over 7000 kWh

The proportion of electricity in terminal energy consumption has increased year by year



Forecast of China's urbanization rate

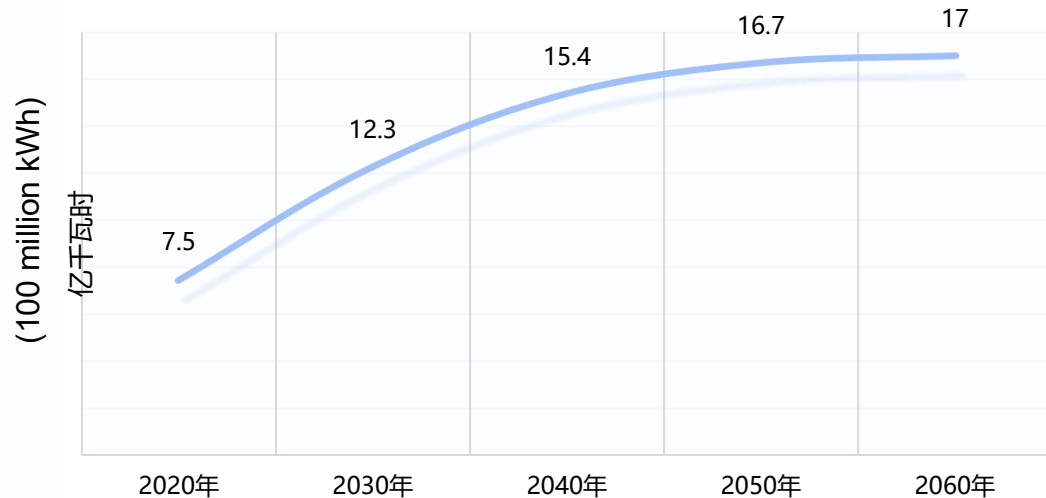


Growth expectations for electricity demand

电力电量需求的增长预期

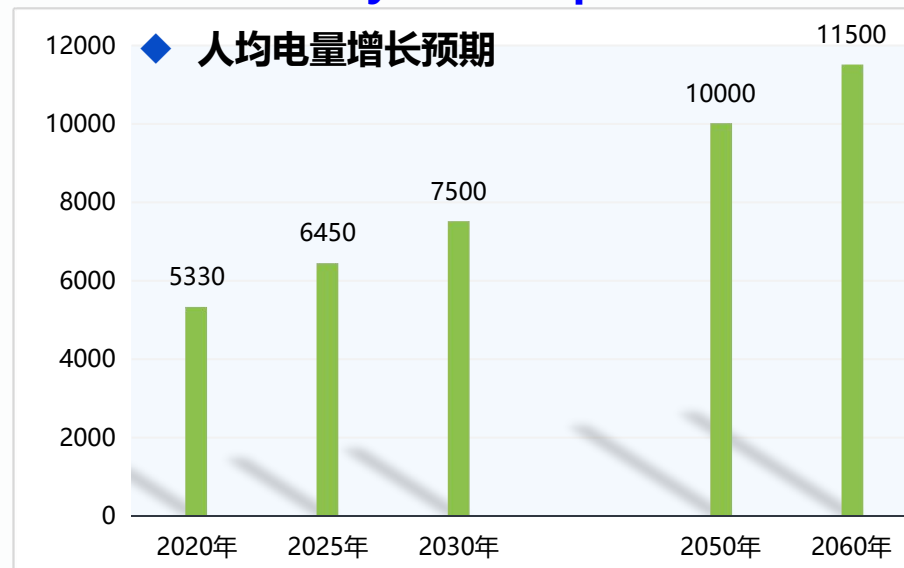
◆ Expected growth in total electricity consumption

◆ 总电量增长预期



◆ Expected growth in per capita electricity consumption

◆ 人均电量增长预期

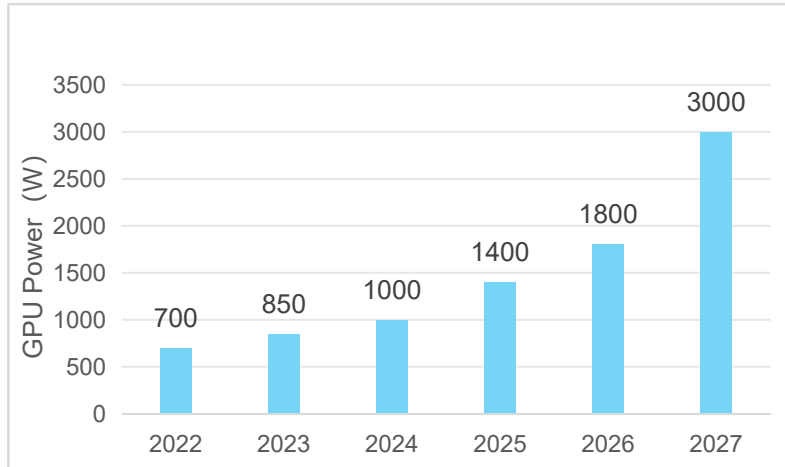


Academician Tang:

- ◆ By the end of 2024, the total electricity consumption was 10 trillion kWhs.
- ◆ West-to-east power transmission capacity exceeds 300 million kilowatts, through cross-provincial and cross-regional channels, transmitted more than 20% of new energy generation
- ◆ It is estimated that by 2060, the total electricity consumption will exceed 20 trillion kWhs, with a significant increase in the scale of West-to-East power transmission, and new energy electricity accounting for about 70% via cross-provincial and cross-regional channels.

AI and data centers have become new global drivers of electricity growth

AI和数据中心成为全球新的电力增长点



The AI industry's share of global power consumption is expected to grow significantly.

According to data from the International Energy Agency (IEA), by 2030, global data center electricity consumption will account for about 4% of the world's total electricity consumption. Especially with the huge demand for computing power from deep learning.

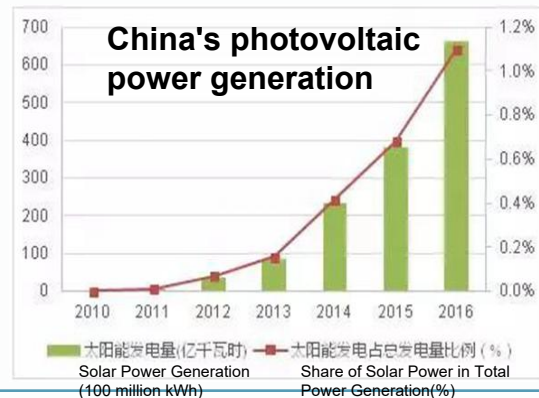
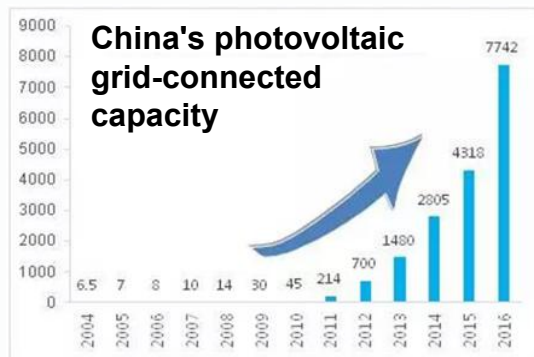
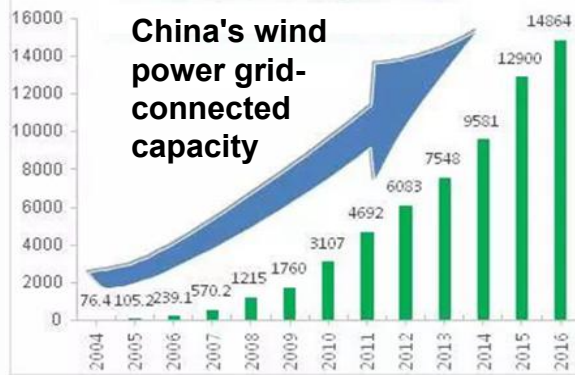
- ◆ **The digital economy is driving a surge in computing power demand.**
- ◆ **Data centers are expanding both in scale and rack numbers.**
- ◆ **Both computing equipment operation and cooling energy consumption are high, running continuously 24 hours a day, with cooling systems accounting for 30%-40% of total energy consumption.**
- ◆ **Energy consumption for data storage and processing volumes have also greatly increased, the global data volume growing at an average annual rate exceeds 20%.**

New Power System Feature- 2: Rapid Growth in New Energy Power Generation Installations

新型电力系统的主要特征-2 新能源发电装机增长迅速

The proportion of new energy power generation installed capacity and power generation is growing rapidly

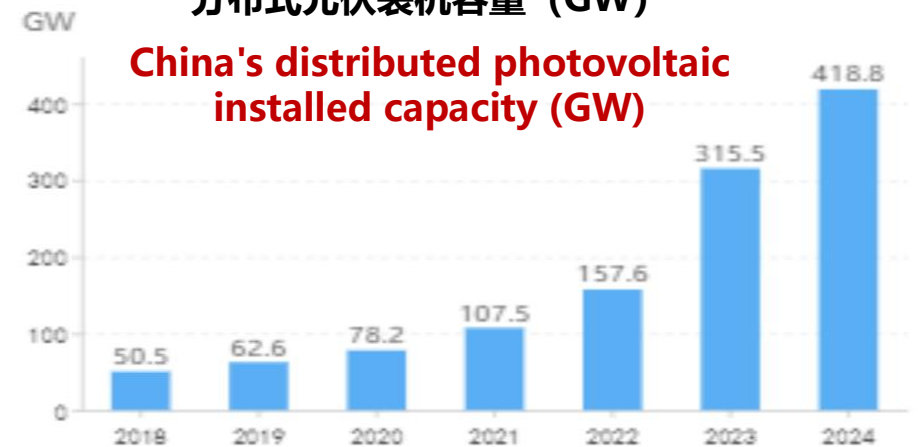
新能源发电装机和发电量比例快速增长



Not only the installed capacity and power generation of centralized wind and PV are growing rapidly, but the installed capacity of distributed PV is also growing quickly. For example, in 2023, Shandong Province's total installed capacity of distributed and residential PV reached 59GW.

不仅集中式风电光伏装机容量和发电量增长迅速，分布式光伏的装机容量也在快速增长。例如，2023年，山东省分布式和户用光伏的装机总容量已达到59GW。

分布式光伏装机容量 (GW)



2020-2030 power generation installed capacity forecast (100 million KW)

2020-2030发电装机预测 (亿kW)

In 2025, the total installed power capacity will reach **3.9 billion KWs**, with wind power accounting for **640 million KWs**, solar power **1.2 billion KWs**, and wind-solar power making up **47%** of the total. Hydropower will be **450 million KWs**, nuclear power **65 million KWs**, while coal-fired power capacity will peak at **1.27 billion KWs**.

2025年，电源总装机达到39亿千瓦，其中风 6.4 亿、光 12 亿、风光占比47%，水电 4.5 亿、核电 0.65 亿千瓦，煤电容量达峰，装机12.7亿千瓦。

In 2025, China's power generation will reach approximately **10.5 trillion kWhs**, with wind and solar power accounting for **2.3 trillion kWhs**, representing **21%** of the total.

2025年，发电量约为10.5万亿kWh，其中风光发电量2.3万亿kWh，占比21%。

By 2030, the total installed power capacity will reach **5.4 billion KWs** with **3.4 billion KWs** from clean energy, accounting for **63%**, and **2.8 billion KWs** from wind and solar power, accounting for **51%**.

2030年，电源总装机将达54亿千瓦，其中清洁能源装机34亿千瓦，占比63%；风、光装机28亿千瓦，占比51%。

By 2030, the expected power generation will exceed **13 trillion kWhs**, with non fossil energy accounting for **50%** and wind and solar power accounting more than **30%**.

2030年，发电量预计超13万亿kWh，其中非化石能源发电量占比达50%，风光发电量占比达30%。

New Power System Feature- 3: Safety Prevention and Emergency Response Capabilities Urgently Need to Improvement

新型电力系统的主要特征-3 安全防范和风险应急能力亟待提高

The structure of the power grid is becoming increasingly complex, threats from natural disasters and the risk of cyberattacks from external forces increases, and safety prevention and emergency response capabilities of power system is urgently need to improve.

电网结构日益复杂，自然灾害的威胁、来自外部势力网络攻击的风险

**Central China Power
Grid Vibration
Accident (2006.7.1)**

**Hainan Power Grid
Province-wide Outage
(2005.11.9)**

In 2008, heavy snow and freezing rain caused severe damage to the power grids of more than ten southern provinces, This caused direct economic losses totaling more than 100 billion yuan.

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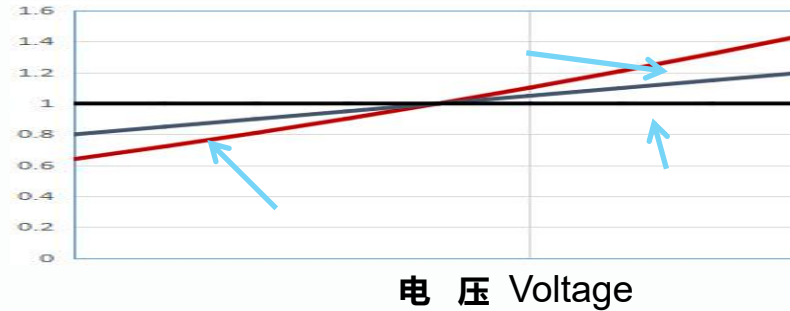
Security prevention and emergency response capabilities urgently need to improvement

安全防范和风险应急能力亟待提高

The proportion of power received outside large cities increases, system inertia levels decrease, source and load balancing capacity decreases, and load characteristics tend to be more constant and power-based,
大城市外受电比例加大、系统惯性水平下降、源荷平衡能力降低、负荷特性更趋恒功率化、... ..

The proportion of electricity received outside the urban cities has increased year by year

$$P = \frac{V^2}{Z_{const}} \Rightarrow P = \text{CONST}$$



Load characteristics are more constant, with more dynamic power consumption



Power electronics leads to a decrease system inertia levels, and the system's ability to balance source-load power fluctuations is continuously weakening.

How can we respond to changes in new energy and load characteristics, prevent large-scale power outages, guard against cyberattacks, and ensure the safe and stable operation of increasingly complex power systems?
如何应对新能源和负荷特性的变化, 防范大面积停电, 防范网络攻击, 确保日益复杂的电力系统安全稳定运行?

New Power System Feature- 4: Users' demands for power supply reliability and power quality are increasing

新型电力系统的主要特征-4 用户对供电可靠性和电能质量要求日益提高

Comparison of power supply reliability

	China	Singapore	Japan
Average power supply reliability %	99.9400	99.9999	99.9996
Average outage duration	5.22h	0.31min	2min

Forecasts for U.S. users' reliability requirements for grid power supply over the next 30 years

Reliability	At present	The future
99.9999%	8%-10%	60%
99.9999999%	0.6%	10%

Distribution network structure

Major cities in China	Ring network structure, radial operation
Paris district	Three-ring network T-connection or double-ring network T-connection
Central London	Multi-branch, multi-connector wiring
Downtown Tokyo	Main line backup wire, ring and point network wiring
Singapore	Closed ring network petal wiring

	China	Developed countries
Cable conversion rate	22.40%	80%+
Distribution automation coverage	20%	90%+
Distribution asset management	Extensive style Management	Refined transformation Efficient management

Compared with developed countries, there is still a significant gap in power supply reliability

与发达国家和地区相比，供电可靠性仍存在较大差距

Users are increasingly demanding power supply reliability and power quality

用户对供电可靠性和电能质量要求日益提高

Digital loads are demanding increasingly high power quality and reliability

数字化负荷对供电质量和可靠性要求越来越高

- ◆ More and more electrical devices are sensitive to voltage drops, with digital economy loads accounting for 13% - 50%
- ◆ Voltage drop causes prolonged shutdowns for sensitive users, with 1-10 CYC disturbances
- ◆ 80% of complaints about power quality issues are about voltage sagging
- ◆ Organically link power supply reliability with power quality indicators.

RS	Average outage duration	Average outage duration*	RS*	Specific gravity%
	min	*(This was caused by a temporary surrender min)		
99.945	288.2	5.71	99.944	1.94
99.99	52.02		99.989	9.89
99.999	5.202		99.998	52.31

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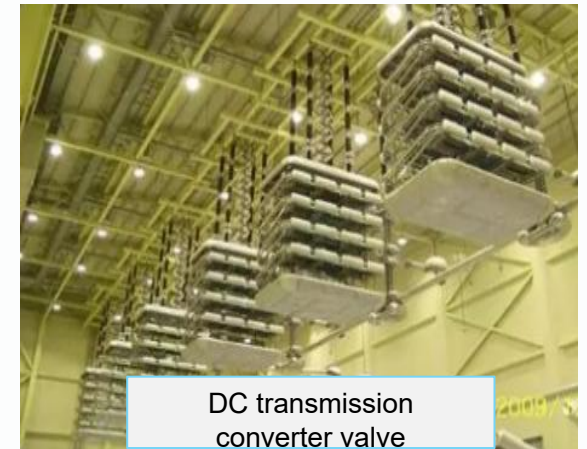
How can we meet users' ever-increasing demands for power supply reliability, power quality, and other services?

如何满足用户对供电可靠性、电能质量等服务的日益提高的要求？

Key features of new power systems—5: The trend toward power electronics in systems is accelerating

新型电力系统的主要特征 - 5 系统的电力电子化趋势加速

- Power electronic devices such as silicon carbide(SiC) achieve higher cost-performance ratios and larger capability
- Flexible DC transmission technology has developed rapidly, with a large number of flexible DC devices being connected to the grid
- Wind, PV and energy storages are connected to the grid through power electronic converters
- Power electronic switches will gradually replace the mechanical switchers and will change the control distribution grid
- Data centers supported by power electronics equipment are booming
- In enterprise production, large amount of power electronic devices is used in flexible production lines, speed control, computer control, etc.



New Power System Feature- 6: Energy storage has become an important means to support new power systems

新型电力系统的主要特征- 6 储能成为支撑新型电力系统重要手段

1. Lithium-ion batteries (smoothing volatility in new energy)

- **High energy density (150-210 Wh/kg) and fast response speed**
- **Safety (focusing on lithium battery safety issues), environmental friendliness, cycle life (5000-10000 cycles)**

2. Flow batteries (3000-7500 RMB/kWh)

- **It offers good safety, long lifespan, and high residual value**
- **Low energy density**

3. Pumped hydro energy storage, compressed air energy storage (smoothing the intermittency of new energy)

- **Pumped storage: Centralized, large capacity, regionally constrained**
- **Compressed air: Long lifespan, greater capacity means better efficiency, not limited by region, low energy density**

4. Storage of Cooling, heat, and hydrogen

5. Other energy storage (flywheel, supercapacitor, gravity、 . . .)

New Power System Feature- 7: Increasing regulation needs at both ends of new power systems

新型电力系统的主要特征-7 新型电力系统的两端调节需求加大

- **The inertia of the power system decreases, and the regulation demand increasing**
 - 电力系统的惯性降低，调节需求加大
- **The distribution system connects to a large amount of distributed energy, the load characteristics has been significantly changed**
 - 配电系统的接入大量分布式能源，负荷特性发生重大变化
- **The electric market needs more and more precise interactions between supply and demand side**
 - 电力市场化进程推动更精准的供需互动

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Market mechanisms need sinteraction and coordination among source-grid-load-storage

市场机制下源-网-荷-储的互动和协调

- Renewable energies are being used by optimal allocation and efficient consumption, and the flexible regulation capacity of the grid is enhanced, under continuously improving market mechanisms, through the interaction and coordination of source, grid, load, and storage.
- Diversified market transactions are conducted continuously, to stimulate enterprise competitiveness, encourage users to participate in auxiliary services, and strengthen coordinated regulation of source, grid, load, and storage.
- 在不断完善的机制下，通过源、网、荷、储**互动协调**，可再生能源**高效利用**和优化配置，提高新能源消纳水平、提高电网**柔性调控能力**。
- 开展**多元化市场交易**，激发企业**竞争活力**，鼓励用户参与**辅助服务**，加强源网荷储的协同调控。



Building a new type of power system

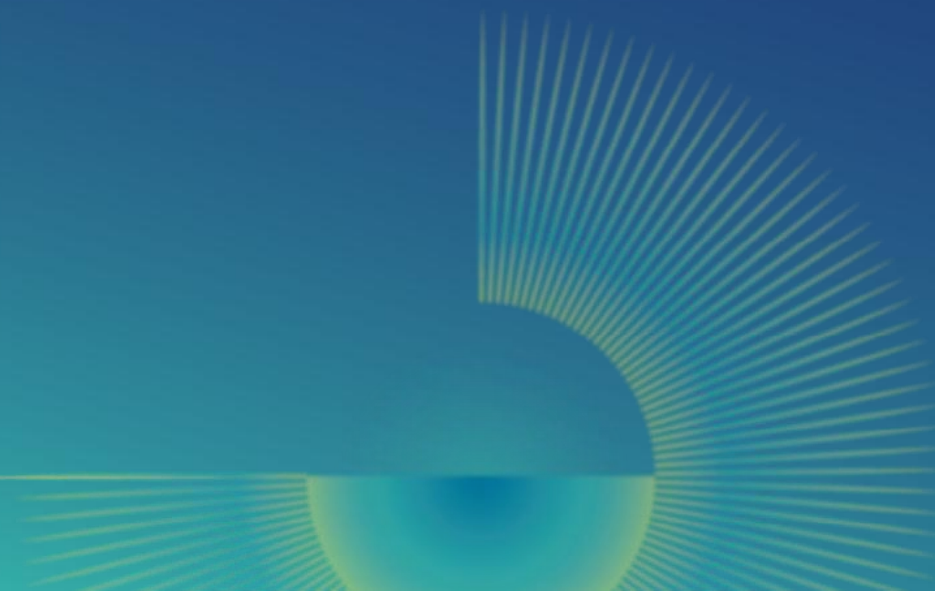
构建新型电力系统

- ◆ **Strategy:** Control the total amount of fossil energy, improve energy utilization efficiency, implement renewable energy substitution, deepen power system reform, build a new power system dominated by new energy, achieve the strategic goals of carbon peak by 2030 and carbon neutrality by 2060.
- ◆ **Technology:** Innovative technologies such as AI, big data, and the Internet of Things have promoted the efficient and reasonable consumption of renewable energy. Various kinds of energies, such as electric cooling and heating, have show the deeper coupling characteristics. Diversified new energy storage technologies have expanded the feasible range of operation in power systems, continuously expanding the space for optimal allocation of energy resources.
 - The urgent need for energy transition and decarbonization -- **necessity**
 - Innovative application of new technologies development – **feasibility**

Become an important foundation for building new power system

TWO

Technical directions for new power systems



The operation of new power systems exhibits new characteristics

新型电力系统运行呈现新的特征

- **Efficient absorption of high proportion of new energy increases regulation demand at both ends of the system.**
 - 高比例新能源的高效消纳，系统两端调节需求增大。
- **A large amount of power electronic equipment is widely accessed, leading to a decline in system inertia.**
 - 大量电力电子装备广泛接入，系统惯性水平下降。
- **The continuous growth in load and the demand for high-quality power supply by users have raised higher requirements for the planning, construction, operation, and digital transformation of distribution networks.**
 - 负荷持续增长和高质量供电的用户需求，对配电网的规划、建设、运行和数智化改造提出了更高的要求。
- **Conventional coal-fired power units are gradually being phased out, challenging the system's stability and balancing capability.**
 - 常规煤电机组逐步退出，挑战系统的稳定和平衡能力。
- ◆ **Focusing on forward-looking and directional innovative technologies supporting new power systems over the next 10-20 years.**
 - ◆ 关注未来10-20年，支撑新型电力系统的前瞻性和方向性的创新技术。

- LLMs are enhancers of agent capabilities, dramatically improving AI agents' understanding and generalization abilities; The degree of interaction and collaboration is an extension of Agent's capabilities, providing a more personalized, coherent, and smooth interaction experience.
- Intelligent brain under multiple agents collaborates with source, grid, load, and storage for optimization is an AI application in new power system dispatching.
- LLM是Agent能力的增效器，使得AI Agent的理解力和泛化能力急剧提升；交互协作程度是Agent能力的扩展器，提供了更个性化、连贯流畅的交互体验。
- 多智能体下的智能大脑与源网荷储协同优化是AI在新型电力系统调度应用的典型案例。
- Based on small-parameter offline large models, and based on power system dispatching procedures and international standards, a professional power dispatching large model can be constructed.
- 基于小参数离线大模型，依据电力系统专业书籍、调度规程、国际技术标准，通过指令微调技术，构建电力调度专业大模型。

Intelligent agent of power system dispatching

电力系统调控应用智能体

- **Overall monitoring agent:** Overall monitoring the flexible resources such as power supply, energy storage, and adjustable loads across the power network
 - **Situational awareness agent:** Performs time-series tracking and situational awareness of the grid's power balance and cross-sections
 - **Collaborative control agent:** Enhances power balance capability by proposing a source-grid-load-storage collaborative frequency control method, to improve effective frequency control of the grid under normal and emergency conditions
 - **Decision Assistance Agent:** Enables early warning of new energy risk of exceeding operational limits within several seconds, and assists decision-making for risk suppress.
-
- **全景监视智能体:** 对全网电源、储能、可调节负荷等灵活性资源进行全景监视;
 - **态势感知智能体:** 对全网电力电量平衡和断面进行时序跟踪与态势感知;
 - **协同控制智能体:** 提升电力电量平衡能力, 提出源网荷储协同频率控制方法, 提升对电网在正常和紧急情况下的有效频率控制能力;
 - **辅助决策智能体:** 实现对新能源越限风险的秒级预警与辅助风险平抑决策。



Prospects for the application of artificial intelligence in electric power

电力人工智能应用前景

1. Generation-grid-load-storage coordination to build a new power system characterized by "active interaction."

源网荷储协同，构建“主动交互”的新型电力系统

Promote the transformation from "source follows load" to "load follows source": enhance new energy absorption capacity. Activate user-side flexible loads to achieve distributed resource collaborative response.

2. Deep integration of market and low-carbon is giving rise to new scenarios of "AI + energy trading."

市场化与低碳深度绑定，催生“AI + 能源交易”新场景

By using algorithms to match electricity price fluctuations and generation output in real time, it helps enterprises accurately identify the optimal trading partners, reduce energy costs, improve the return rate of new energy projects, and support the industry's carbon reduction.

3. Dual empowerment between electricity and computing power, supporting the global AI competition landscape

电力 - 算力双向赋能，支撑全球 AI 竞赛格局

Ample power has become the core support for AI computing power expansion, while AI in turn optimizes power resource allocation, this forms a collaborative ecology system of "green power - computing power - applications."

4.

Artificial intelligence has become an important technological support for new power systems

人工智能成为新型电力系统的重要技术支撑

Application cases of power artificial intelligence

电力人工智能应用案例

1. **Intelligent operation and maintenance of high-voltage equipment:** robots replace manual labor for efficient operation 高压设备智能运维：机器人替代人工高效作业
 2. **AI Scheduling and Forecasting:** AI helps to solve the consumption of new energy 调度与预测：AI 破解新能源消纳难题
 3. **Power grid emergency and repair:** Unmanned operation to make safty assurance throughout the entire process 电网应急与抢修：全流程无人化提速保供
 4.
- ◆ The integration of AI with advanced sensing, the Internet of Things, big data, cloud computing, and other information and communication technologies, along with the application of 5G and BeiDou technology in the power grid, will revolutionize traditional power system planning, operation, management, dispatching, control, and transaction methods.

- ◆ 人工智能与先进传感、物联网、大数据、云计算等信息通信技术融合，5G和北斗技术在电网的应用，将颠覆传统电力系统规划、运行、管理、调度、控制和交易方式。

Applications of power big data and knowledge graphs

电力大数据与知识图谱应用

Pain points encountered in AI application of power systems

- **Data lakes have become data swamps**
- Digital twins have become 3D displays

The "Illusion" of Big Language

The data middle platform has turned into a data swamp
数据中台变为数据沼泽

The digital twin of the power grid has degenerated into a 3D
visualization
电网数字孪生沦为3D展示

The effectiveness of AI projects is not ideal
人工智能项目效果不理想

The digital world calls for a foundation of digital knowledge

数字世界呼唤 数字知识底座

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Power system business data is decentralized, integrating power data resources for global collaborative application

电力系统业务数据分散，整合电力数据资源进行全局协同应用

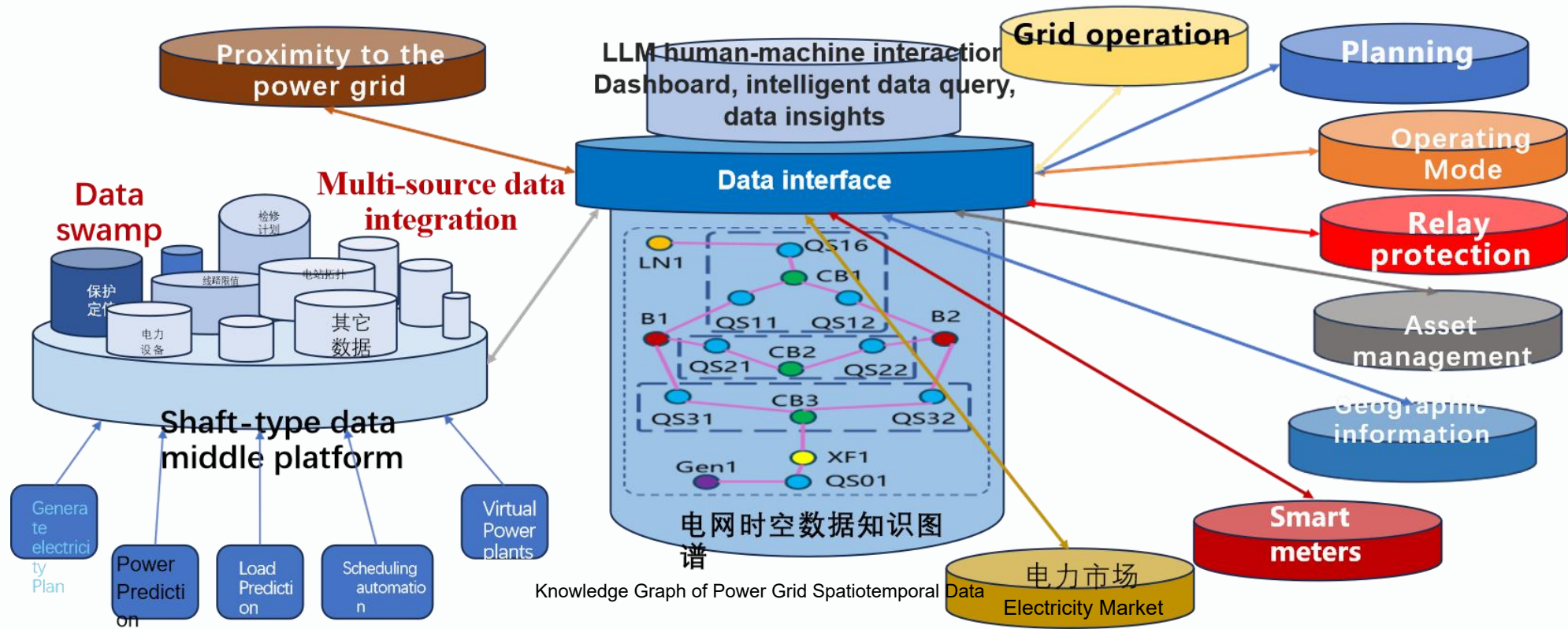
- Each business domain in power utility is modeled independently with fragmented calculations, resulting in insufficient value realization;
 - Inefficient data access for massive amounts of data, and there is no efficient data sharing among departments
 - As fluctuations in source-grid-load-storage cause rapid changes in a short period, there are bottlenecks in computational efficiency in the fields of risk prediction and global resource coordination.
- 各业务域独立建模，分区域割裂计算，价值发挥不充分；
 - 常规关系型数据组织面对海量数据存取效率低下，高效共享能力不足；
 - 源网荷储波动导致短时间风险变化迅速，风险预知和资源全局协同都存在计算效率瓶颈。

Business data fragmentation has become the main challenge, data lacks efficient organization, and global resource collaborative analysis is insufficient

业务数据割裂成为主要难点，时空数据缺乏高效组织，资源全局协同分析不足

Break through the "data swamp," achieve multi-source data integration, and build a digital foundation for AI

打破“数据沼泽”，实现多源数据融合，构建人工智能数字底座

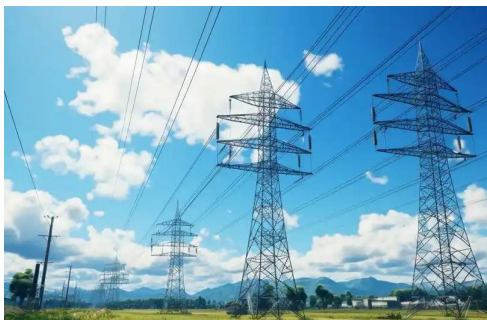


Build a data map of the power grid integrating multi-source data, supporting multiple services in using a single data view

构建集成多源数据的电网时空数据图谱，支撑多元业务使用同一数据视图

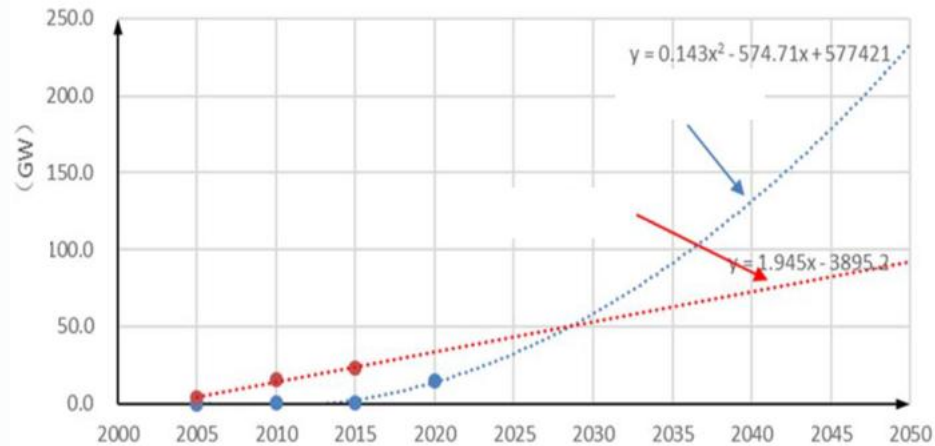
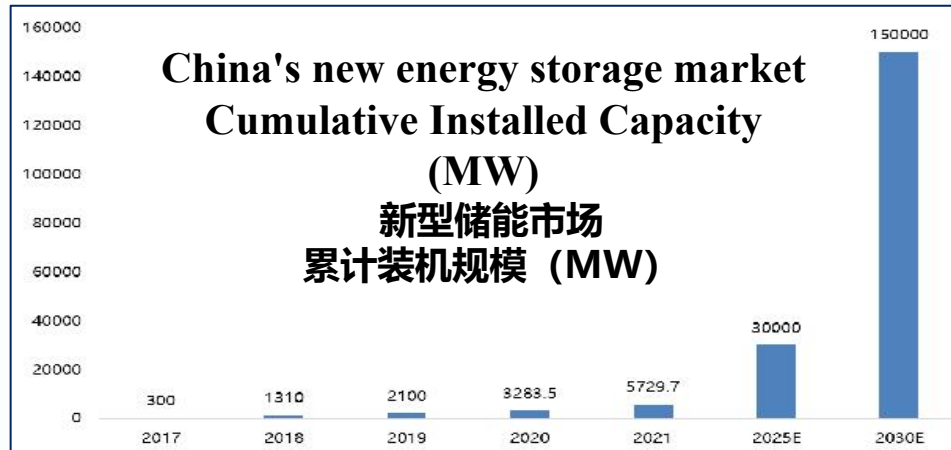
智能电网技术 Smart Grid technologies

- ◆ 在机电系统建模基础上，对海量电力电子装备、多元负荷等构成复杂巨系统建模，耦合电力气象预测，增强仿真能力，通过预判、预警、预控，实现安全风险主动防御。
- ◆ Complex giant systems with massive power electronic equipment are modeled, to enhance simulation and analysis capabilities, and realize active defense of security risks through prediction, early warning and pre-control.
- ◆ 发展特高压、柔性输电、直流电网等先进输电技术，探索高温超导等新型输电技术。
- ◆ Develop advanced technologies such as ultra-high voltage transmission and flexible DC power grid. And to explore new transmission technologies such as superconductivity.
- ◆ 突破主动配电网、直流配用电、智能柔性配电、微电网运行调节等关键技术。
- ◆ Make breakthroughs in key technologies such as active distribution network (ADN), DC and intelligent flexible power distribution, microgrid operation control.



Frontier technology of the new power system 2 -- Efficient, high-density, low-cost, and long-life energy storage

新型电力系统的技术方向之二 --- 高效、高密度、低成本、长寿命储能



The application of energy storage technology runs through every stage of the new power system's generation, transmission, distribution, and usage.

储能技术的应用贯穿于新型电力系统发、输、配、用的各个环节。

Diversified energy storage technologies such as electric storage, thermal storage, gas storage, and hydrogen storage, break through the bottleneck of the need for real-time balancing of power system generation, transmission, distribution, and usage, supporting multi-energy interconnection and multi-grid integration, and become one of the disruptive technologies that will change traditional power system architecture and operating models.

储电、储热、储气、储氢等多元化储能技术打破了电力系统发输配用必须实时平衡的瓶颈，支撑了多能互联互通和多网融合，有可能成为改变传统电力系统架构和运营模式的颠覆性技术之一。

Grid-forming energy storage and grid-following energy storage

构网型储能与跟网型储能

Grid-following energy storage	Grid-forming energy storage
◆ The current source generally uses PQ constant power control ◆ Phase Locked Loops (PLLs) track grid phases ◆ Passively performs charge-discharge operation according to instructions ◆ Inertia cannot be supplied to the system ◆ Simple control, low cost, mature and stable ◆ Easily disconnected when there is no network or weak connection ◆ Suitable for stable grid areas, peak shaving and valley filling for commerce and industry, smoothing wind and solar fluctuations, and power market arbitrage	◆ The voltage source generally uses VF constant voltage frequency control ◆ It can independently establish and maintain voltage and frequency ◆ Actively supports grid-side frequency and voltage regulation, with black start capability ◆ Synchronous generator inertia (Virtual Synchronous Machine VSG) can be provided. ◆ Control is complex and costly ◆ It can operate independently in isolated islands or off-grid ◆ Suitable for large new energy bases, weak current grids, island microgrids, and scenarios requiring black start and frequency regulation and voltage regulation support

储能技术 Energy storage

- ◆ 突破电化学储能热稳定性、系统集成、梯次利用、纳米材料等关键技术，研发钠离子、钾离子等替代技术。
- ◆ Speed up the development of key technologies in electrochemical energy storage, including thermal stability, system integration, cascade utilization, nano materials, etc. Research on sodium and potassium storage technologies.
- ◆ 推动压缩空气储能、飞轮储能、重力储能等技术向大容量、高效率、灵活化等方向发展。
- ◆ Promote the development of compressed air energy storage, flywheel energy storage, gravity energy storage and other technologies for large capacity, high efficiency and flexibility.
- ◆ 发展“煤电+”、熔盐储热、氢储能等长时储能技术。
- ◆ Develop long-term energy storage technologies such as coal power plant plus molten salt heat storage, and hydrogen storage, etc.
- ◆ 突破质子交换膜和高温固体氧化物电解制氢等关键技术。
- ◆ Make breakthroughs in key technologies such as proton exchange membranes and high-temperature solid oxide electrolysis for hydrogen production.
- ◆ 加强氢脆失效、低温液态储存、固态金属储存等方面基础研究，推进氢内燃机、燃料电池和氢冶金等关键技术研发。
- ◆ Strengthen research on hydrogen embrittlement failure, low-temperature liquid storage, solid metal storage, etc. Promote the research and development of key technologies such as hydrogen internal combustion engines, fuel cells and hydrogen metallurgy.



Frontier technology of New Power Systems 3--- Highly reliable, low-loss power electronic equipment and technologies

新型电力系统的技术方向之三 --- 高可靠低损耗的电力电子装备和技术



Power electronic devices are suitable for large new energy converters, medium- and low-voltage distribution networks, or microgrids, bringing revolutionary changes to new power systems.

电力电子器件适用于大型新能源变流装置、中低压配电网或微电网，为新型电力系统带来革命性变化。

- **Device upgrades:** Silicon carbide (SiC), gallium nitride (GaN), and other devices are gradually replacing traditional silicon devices, providing advantages such as high voltage resistance, low loss, and fast switching speeds, with high efficiency, energy saving, and miniaturization.
- **器件升级:** 碳化硅 (SiC)、氮化镓 (GaN) 等器件逐步取代传统硅器件，具有高耐压、低损耗、开关速度快的优势，推动电力电子系统向高效节能、小型化方向发展。
- **Performance breakthroughs:** Toward high voltage and large capacity, meeting the new power system's needs for large-scale energy transmission and distributed resource collaboration.
- **性能突破:** 朝高电压、大容量方向演进，满足新型电力系统对大规模能源传输、分布式资源协同的需求，支撑跨区域电网互联。
- **Collaboration with AI and digital twins:** Integrates self-diagnosis and adaptive functions, combined with AI algorithms for dynamic optimization of operating conditions, and pairs with digital twin technology to complete virtual simulation and control, improving operational efficiency and reliability.
- **与 AI、数字孪生协同:** 集成自诊断、自适应功能，结合 AI 算法实现工况动态优化，搭配数字孪生技术完成虚拟仿真与实景调控，提升运维效率和可靠性。

Highly reliable, low-loss power electronic equipment and technologies

高可靠低损耗的电力电子装备和技术

Application of power electronics technology in electrical power

- **Grid-side - Flexibility and Stability Control:** Flexible AC transmission systems (FACTS), STATCOM (Static Synchronous Compensator), SVC (Static Reactive Power Compensator) and SVG, etc.
 - **Power generation side - efficient grid integration of new energy:** convert frequency and voltage of electric energy, meeting grid connection requirements, suppressing fluctuations in wind and solar power generation.
 - **Energy Storage and User Side - Flexible Energy Optimization:** efficient charging and discharge conversion, supporting load regulation for grid "off-peak charging and peak discharge".
 - **Distribution side - intelligent upgrade:** Power electronic devices in the distribution network (such as smart inverters) support distributed generation access, combined with 5G communication for real-time monitoring and control, enhancing the reliability and flexibility of the distribution network.
-
- **Power electronics devices provide flexible control interfaces for reliably connecting distributed energy to new power systems.**
 - **High-reliability, low-loss power electronics technology has formed active, flexible, and controllable AC-DC hybrid transmission and distribution networks, changing the control and operation modes of traditional transmission and distribution networks.**
 - **Characteristic testing and operational status monitoring of power electronic devices will become new market demands.**



High efficiency and low-cost wind power generation

The single unit capacity of wind power generation is being larger, and the output power is continuously improved.

- ❑ Longer and lighter blades
- ❑ Modular blade
- ❑ Integrated drive chain
- ❑ Control system
- ❑ Laser radar wind measurement
- ❑ New turbine shroud
- ❑ More optimized wind turbine tower
- ❑ Silicon carbide frequency converter
- ❑ Superconducting generator

Growth trend of wind turbine capacity and wind turbine diameter

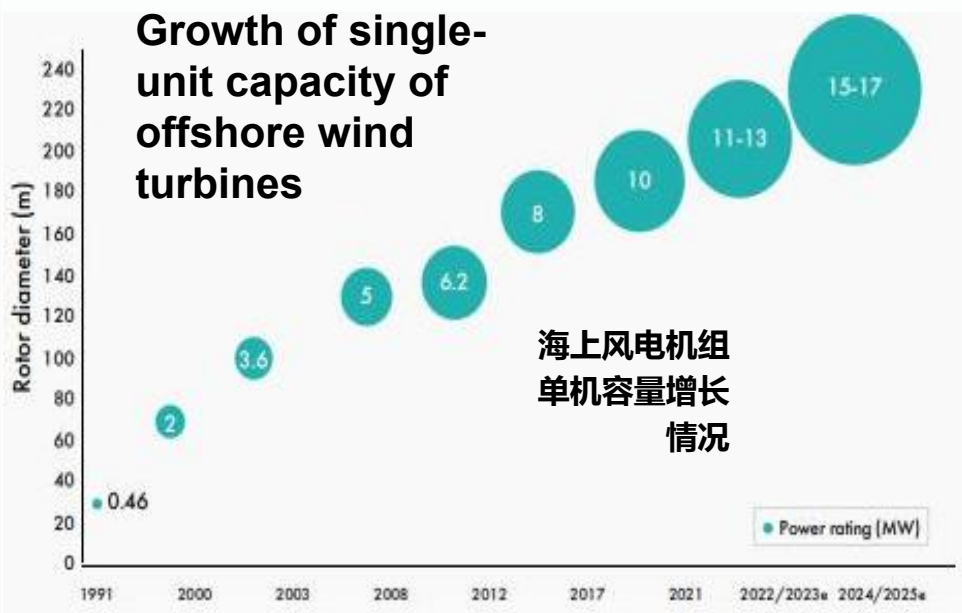
Large-scale wind power generation technology

大型风力发电技术

	Large-scale onshore wind power	Large offshore wind power
Environmental characteristics	Wind speed fluctuation is large, terrain is complex (mountainous/plain), and there are noise and visual impact limitations	Stable wind speed (annual utilization hours 2500-4000h, higher than 1500-2500h on land), high humidity, seawater corrosion, strong typhoons/giant waves
Core technical challenges	1. Suitable for low wind speed areas (high tower + long blades); 2. Noise reduction design ; 3. Terrain adaptability (flexible towers, split foundations).	1. Corrosion resistance technology (submarine cables and unit components use zinc-aluminum coating); 2. Typhoon-resistant design; 3. Offshore operations and maintenance (unmanned inspection, helicopter/operation and maintenance vessel dispatch).
Key supporting technologies	Land-based collection lines (mainly overhead lines), substations	Submarine cables, offshore booster stations (modular design, easy installation)
Cost structure	Machinery costs account for about 60%, with low operation and maintenance costs (about 0.02 RMB/kWh)	Foundation and submarine cable costs account for 40%-50%, with high operation and maintenance costs (about 0.05-0.08 RMB/kWh)

In 2024, the average single-unit capacity of China's offshore wind turbines is 17MW, with an average turbine diameter of 271.5m. The maximum single-unit capacity reaches 26MW. Offshore wind turbines of 10 MW and above account for 58%.

2024 年，中国海上风电机组平均单机容量为17.57MW，平均风轮直径为271.5m。最大单机容量达到26MW。10 兆瓦及以上海上风电机组占比达 58%。



High efficiency and low-cost solar and wind power generation

高效低成本的风光发电技术

High efficient and low-cost solar power generation

◆ The conversion efficiency of PV power generation continues to improve

- Multi junction battery 31%-46%
- Crystalline silicon battery 21%-28%
- Single junction gallium arsenide battery 27% -30%
- Thin film technology battery 14%-24%
- Other new types of photovoltaic cells 10%-22%



◆ The efficiency of solar thermal conversion continues to improve

□ Slot style:

- Improvement of Thermal Fluid
- Efficiency improvement of steam turbine
- Improved thermal storage performance
- No longer need heat exchanger

□ Tower type:

- Improvement of Reflectivity and Receiver Efficiency of Heliostat
- Improved overall efficiency of power generation modules and power stations
- The demand for absolute reflection area decreases



清洁发电技术 Clean generation

- ◆ 风电: 发展风功率精准预测、主动构网型风机、叶片回收循环利用、超导风机等技术。海上风电突破直流集电、平台轻量化、 柔性直流组网等技术。
- ◆ For wind power, develop technologies including accurate prediction of wind power, grid-friendly wind turbines, blade recycling, superconducting wind turbines. Make breakthroughs in the DC collector, the lightweight of off-shore wind power platforms, VSC-HVDC technologies, etc.
- ◆ 太阳能发电: 突破低成本高效率光伏发电、建筑光伏 一体化、光伏组件回收再利用等技术。研发新型传热介质, 提高光热发电效率。
- ◆ For PV power, achieve breakthroughs in low-cost and high-ficiency PV power generation, building integrated PV, PV modules recycling, enhance the efficiency of solar thermal power generation, etc.
- ◆ 核电: 加快第三代压水堆、高温气冷堆等技术推广, 推动快堆、模块化小型堆、海上浮动堆等技术应用示范。
- ◆ For nuclear power, promote technologies such as third-generation reactors and high-temperature gas-cooled reactors, promote the demonstration of fast reactors, small modular reactors and offshore floating reactors.

- ◆ Flexible DC transmission, based on voltage source converters, self-shutdown devices, and pulse width modulation (PWM) technology, can supply power to passive networks, there will be no commutation failure, no communication between converter stations, and it is easy to form a multi terminal DC system.
- ◆ 柔性直流输电，以电压源换流器、自关断器件和脉宽调制（PWM）技术为基础，可向无源网络供电、不会出现换相失败、换流站间无需通信、易于构成多端直流系统。
- ◆ New environmental friendly gas insulated high-voltage, high-current, long-distance pipeline transmission technology offers significant advantages such as large transmission capacity, low losses, high reliability, low maintenance, long lifespan, and minimal environmental impact.
- ◆ 新型环保气体绝缘的高压、大电流、长距离管道输电技术，具有输电容量大、损耗小、占地少、可靠性高、维护量小、寿命长、环境影响小等显著优点。

Frontier technology of New Power System 6 - small-scale and safe nuclear power station

新型电力系统的技术方向之六 -- 小型安全的核动力能源站

Thorium-based Molten Salt Experimental Reactor (TMSR) - China's fourth-generation advanced nuclear fission system

钍基熔盐实验堆 (TMSR) - 第四代先进核裂变系统

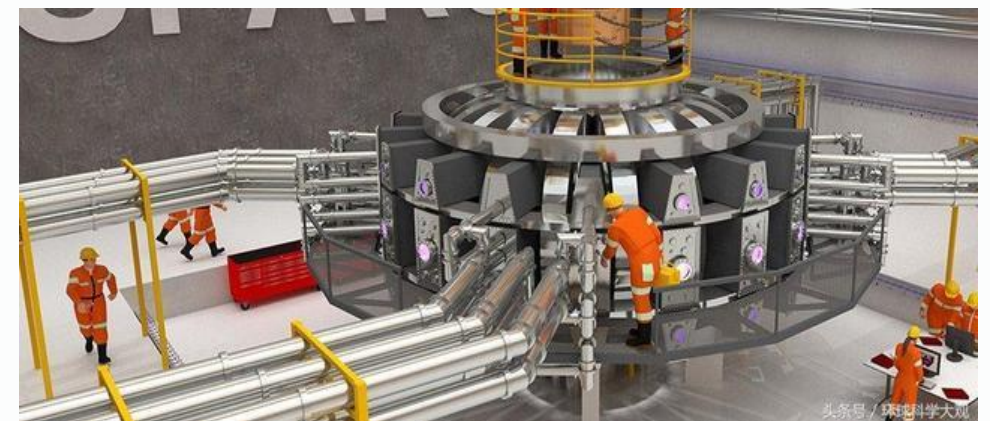
- ◆ **Outstanding safety performance:** Operating at atmospheric pressure with no risk of explosion.
 - ◆ **Abundant resource reserves:** Support energy demand for thousands of years.
 - ◆ **Environmentally friendly and non-proliferated:** nuclear waste is only one-tenth that of uranium piles, and making it difficult to be used in nuclear weapons.
 - ◆ **Multi-scenario adaptation:** Operating temperature around 700°C, requiring no large water sources, suitable for arid regions, and can also link with high-temperature hydrogen production, energy storage, chemical industries, and other industries.
 - ◆ The 2-megawatt experimental reactor in Gansu is currently the only molten salt reactor, with 100% independently controllable in core equipment.
- **安全性能突出:** 常压运行无爆炸风险。
 - **资源储备丰富:** 可支撑数千年能源需求。
 - **环保且防扩散:** 核废料量仅为铀堆的1/10, 难以用于核武器制造。
 - **多场景适配:** 运行温度700°C左右, 无需大量水源, 适配干旱地区, 还可联动高温制氢、储能、化工等产业形成低碳复合系统。
 - 甘肃的2兆瓦实验堆是目前唯一运行并实现钍铀转换的熔盐堆, 核心设备100%自主可控。

Small nuclear fusion technology is also in a rapid development phase, making significant progress in facility construction and financial support.

小型核聚变技术也处于快速发展阶段，在装置建设、资金支持等方面都取得了重要进展

Small nuclear power stations can provide electricity, heat and cool energies for industrial parks.

小型核动力能源站可为工业园区提供电冷热能源。



Frontier technology of New Power System 6 - small-scale and safe nuclear power station

新型电力系统的技术方向之六 -- 小型安全的核动力能源站

Facility construction: A compact fusion energy experimental facility (BEST) is being constructed, expected to be completed by 2027, with a fusion engineering demonstration reactor planned for 2035, and commercial fusion energy generation before 2050.

Capital investment: The fusion research utilities frequently receive startup fundings in recent years. The linear controlled nuclear fusion technology has completed the first round funding of 100 million yuan in 2024.

Diversified Commercialization Models: Before fusion power generation is realized, medical and industrial neutron source technologies such as cancer treatment and nuclear drug isotope production have already made commercial applications.

Stronger policy support: The government has issued strengthen policies to support breakthroughs in key technologies for future energy represented by nuclear fusion and creating a globally influential nuclear power industry cluster.

- **装置建设:** 正在建设紧凑型聚变能实验装置 (BEST), 预计将于 2027 年建成, 2035 年建成聚变工程示范堆, 2050 年前实现聚变能商业化发电。
- **资本投入力度:** 核聚变领域的初创企业频繁获得融资。直线型可控核聚变技术完成亿元天使轮融资。
- **商业化模式多元化:** 聚变衍生技术的产品化和商业化, 计划在聚变发电实现前, 利用聚变中产生的中子开发医疗和工业中子源技术, 服务于癌症治疗、核药同位素生产等领域。
- **政策支持力度增强:** 明确提出要加强以核聚变为代表的未来能源关键核心技术攻关。打造全球有影响力的核电产业集群。



Nuclear power plant will also develop fast

规模化核发电厂也将快速发展

- ◆ Large-scale nuclear power plants are constructed continuously, promoting the comprehensive applications of nuclear power on co-generation of electricity, steam/heat supply, and also on industrial hydrogen production, etc.
- ◆ 核电持续推进基地型规模化发展，推动核能在供汽供热、工业制氢、海水淡化等领域综合利用。
- ◆ To make nuclear power regulation more flexible, and strengthen peak regulation capacity from nuclear power stations.
- ◆ 提升核电调节灵活性，加强调峰能力建设。
- ◆ By 2030 and 2060, the installed capacity of nuclear power generation will reach 120 GW and 400 GW.
- ◆ 预计2030、2060年，核电装机分别达到1.2亿、4亿千瓦以上。

Integrated energy system planning and optimization configuration

综合能源系统规划和优化配置

- ◆ **Analyze and forecast various energy demands**
- ◆ **Based on local resource endowments, optimize the allocation of energy sources**
- ◆ **Analyze the dynamic processes of energy sources and design coordinated control strategies**
- ◆ **Integrated energy system dynamic planning and optimization schemes**

Evaluation of technical economic suitability and cost-benefit analysis of integrated energy systems

综合能源系统的技术经济适用性评价及成本效益分析

- ◆ **Establish models of typical energy and power technology operating and energy consumption characteristics**
- ◆ **Establish evaluation indicators based on energy endowment characteristics, environmental resources, user preferences, and other factors.**
- ◆ **Considering constraints such as market environment, economic impact, and policy changes, an optimization plan for energy and power technology is constructed.**
- ◆ **Conduct cost-benefit analysis and design a business model that is technology suitable and economy feasible.**

Frontier technology of New Power System 7 - Optimized Operation of Integrated Energy Systems

新型电力系统的技术方向之七 --- 综合能源系统的优化运行

Operation control and optimized scheduling of integrated energy systems

综合能源系统的运行控制与优化调度

- ◆ It takes into account the complex coupling of market transactions and different operating entities in optimizing safe operation and scheduling of multiple types of energy networks.
- ◆ Energy flow calculation, safe operation verification, and multi-spatiotemporal dynamic scheduling optimization for multi-energy systems.
- ◆ Integrated energy system coordinated control strategies.

Business models for integrated energy systems

综合能源系统的商业模式

- ◆ Design optimal dynamic allocation plans for marketing resources based on the marketing characteristics of different types of energy services.
- ◆ Guide diversified loads in energy trading, clean energy green certificate trading agency, and design technically applicable and economically feasible business models.

- User-centered value creation
Value-added information centered on data

- Technology-driven business innovation
Leverage reform as an opportunity to tap into benefits

Frontier technology of New Power System 8 -- Market-Driven Optimization of Energy Resource Allocation and Regulation Operation

电网发展技术方向之八 -- 市场驱动的能量资源优化配置和调控运营

健全多层次统一电力市场。推动新能源全面参与市场，探索容量补偿机制、容量市场等多种方式，完善需求侧响应机制。

◆ **Improve the multi-level electricity markets. For promote the full participation of renewable energy, explore capacity compensation mechanisms, capacity markets, improve the response mechanism of demand-side resources.**

加强电碳市场协同。完善碳价传导机制，加强碳市场与绿色电力、绿证市场衔接，充分体现减排贡献的市场价值。

◆ **Strengthen colabration of elerctricity-carbon market, improve the carbon price mechanism, strengthen the connection between the carbon, green electricity and green certificate markets to reflects the market value of emission reduction contribution.**

Figure
A unified electricity market



Figure
Market-oriented operation of integrated energy systems

Market-driven optimization of energy resource operational regulation

市场驱动的能源资源优化配置和调控运营

Improve the electricity price formation mechanism

完善电价形成机制

- ◆ Capacity value 容量价值
- ◆ Electric Energy Value 电能量价值
- ◆ Transmission and distribution segment 输配电环节

Market supervision and policy guidance

市场监管与政策引导

- ◆ Market supervisin 市场监管
- ◆ Subsidy Policies 补贴政策
- ◆ Quota system 配额制度

Promote market integration

推进市场融合

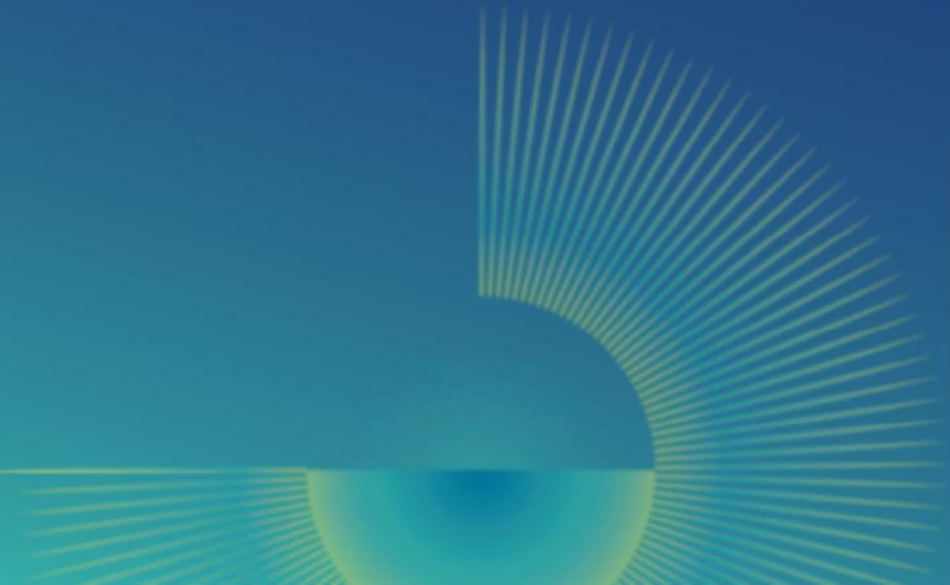
- ◆ Carbon Emission Market 碳排放市场
- ◆ Green certificate trading market 绿证交易市场

Figure

The divergence between electricity market value and price

THREE

**Key application technologies for new
power systems**



Key application technologies in the operation of new power systems

新型电力系统运行中的关键应用技术

To realize the "multi-energy collaborative" operation model and "equal sharing" market transactions in energy and power services, a series of key application technologies must be broken through to support the implementation of the energy and power service.

为了实现能源电力服务“多能协同”的运行模式与“对等共享”的市场交易，需突破一系列关键应用技术，支撑能源电力服务的概念落地。

- ◆ Forecasting distributed energy and various types of energy loads
- ◆ Planning and optimization of new power systems
- ◆ Operation control and optimized dispatching of new power systems
- ◆ Intelligent demand response with interaction — source-grid-storage-load
- ◆ User energy efficiency, customized energy usage, grid auxiliary services, and other application services
- ◆ Situational awareness, non-intrusive measurement, and IoT technologies
- ◆ The market's operating trading mechanisms and techniques
- ◆

Key technologies 1: distributed energy and multi-type energy load forecasting

新型电力系统关键技术之一：分布式能源和多类型用能负荷预测

- High-precision energy production and various energy load forecasts can significantly improve market operational efficiency and enhance the efficiency of energy networks.
- 高精度的能源生产和各类用能负荷预测能够显著提升市场的运行效率，提升能源局域网的运行效率。
- Compared with traditional generation and load forecasting, distributed and diversified energy load forecasting has finer temporal and spatial granularity, is more affected by external factors, and is more challenging.
- 分布式能源和多元化用能负荷预测的时间和空间颗粒度更细，受外部影响的程度更大，难度更高。
- Combine external environmental monitoring with non-intrusive measurements to provide more physical, environmental, and social information to improve prediction accuracy.
- 结合外部环境监测以及非侵入测量提供更多的物理信息、环境信息和社会信息，提升预测精度。
- Artificial intelligence and big data algorithms provide important technical means for precise forecasting of source and load.
- 人工智能和大数据算法为源荷精准预测提供了重要技术手段。

Key technologies 2: **planning and optimized configuration of new power systems**

新型电力系统关键技术之二：新型电力系统规划和优化配置

Planning and optimizing the configuration of new power systems to achieve mutual conversion between "electric-heat" energy sources.

新型电力系统规划和优化配置，实现“电气热”能源间相互转换。

- ◆ Analyze and forecast various energy demands
- ◆ Based on local resource endowments, optimize the allocation of energy hubs
- ◆ Analyze the dynamic processes of energy hubs and design coordinated control strategies
- ◆ Dynamic planning and optimization schemes for new power systems

Evaluation of technical economic applicability and cost-benefit analysis

技术经济适用性评价及成本效益分析

- ◆ Establish models of typical energy technologies such as solar power generation, heat pumps, electric heat storage, electric vehicles, and combined cooling, heating, and power
- ◆ Analyze factors affecting the economic applicability of energy and power technologies, and establish evaluation indicators.
- ◆ Considering constraints such as market environment, economic impact, and policy changes, an energy and power technology optimization plan is constructed.
- ◆ Conduct cost-benefit analysis and design a business model that is technically suitable and economically viable.

Key technologies 3: Operation Control and Optimized Scheduling of New Power Systems

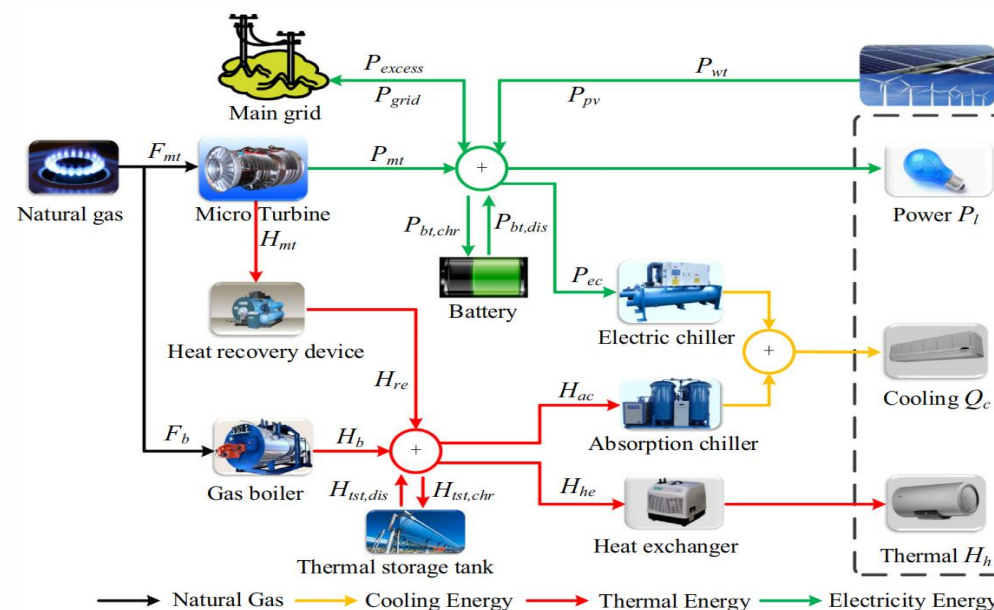
新型电力系统关键技术之三：新型电力系统的运行控制与优化调度

- The operation control and optimized dispatch of new power systems involve complex coupling of multiple energy networks such as electricity, gas, and heat, and must consider the impact of market transactions and different operators on safe operation and scheduling optimization.

新型电力系统的运行控制与优化调度涉及到电、气、热等多类能源网络的复杂耦合，并需要考虑市场交易和不同运行主体对安全运行和调度优化所造成的影响。

- Energy flow calculation, safe operation verification, and multi-spatiotemporal scale scheduling optimization in multi-energy systems have become hot topics in integrated energy regulation.

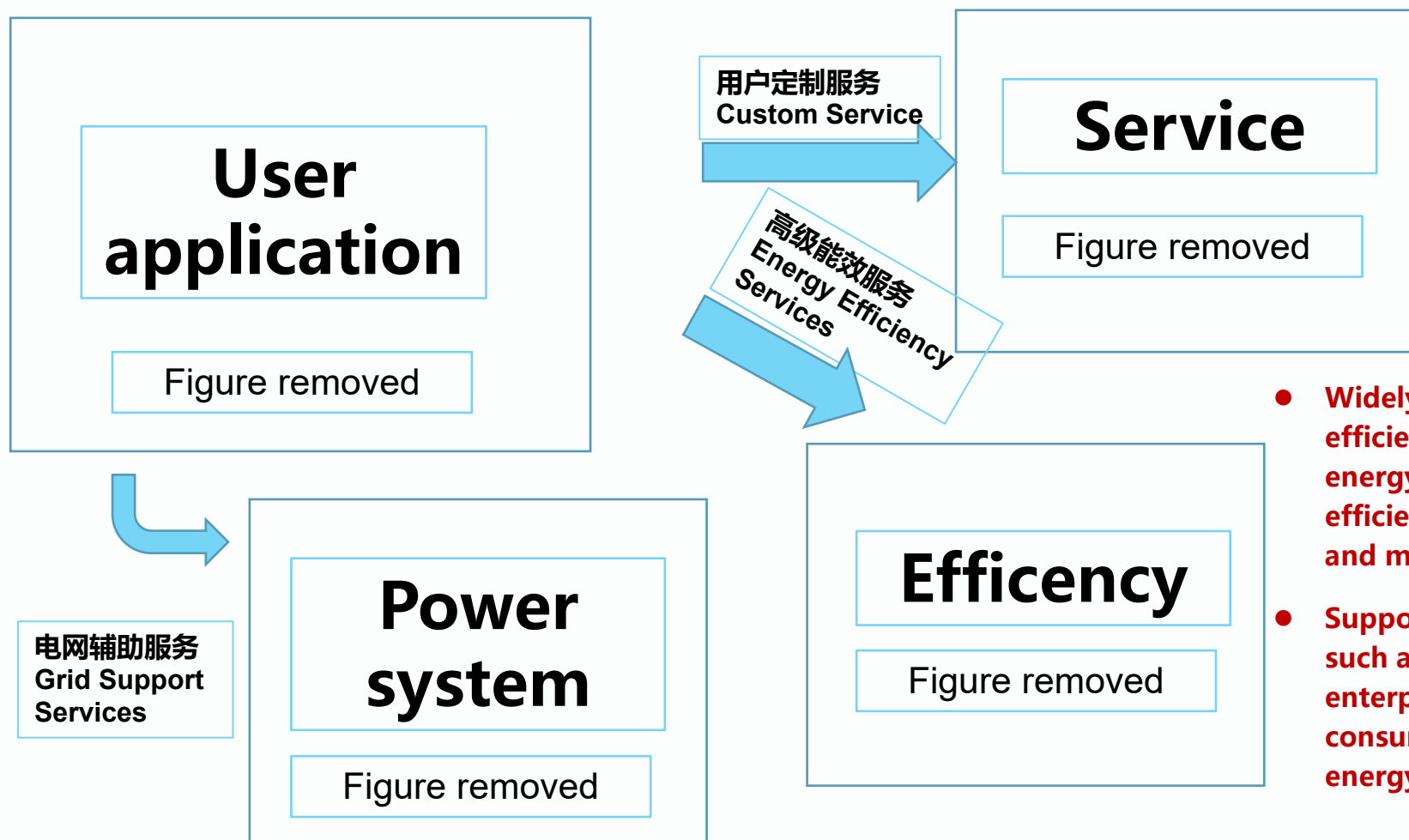
多能源系统的能流计算、安全运行校验与多时空尺度调度优化，成为目前综合能源调控的关注热点。



Zhao Luo, et al: A two-stage optimization and control for CCHP microgrid energy management

Key technologies-4: Advanced application services

新型电力系统关键技术之四：高级应用服务



- Widely pool resources to provide energy efficiency services, and provide customers with energy efficiency services such as energy efficiency management, intelligent operation and maintenance, demand response, etc.
- Support the application of typical scenarios such as commercial buildings, industrial enterprises and parks, reduce energy consumption and improve the comprehensive energy efficiency of the whole society.

Design of comprehensive demand response scheme for multi energy collaboration

面向多能源协同的综合需求响应方案设计

- ① 采集供能信息，接收市场信息
- ② 接受波峰消息，下达降负指令
- ③ 参与需求响应，降低波峰能耗
- ④ 调度能源资源，监控调峰指标

- ① Collect energy supply information and receive market information
- ② Accept the peak message and issue the load reduction instruction
- ③ Participate in demand response and reduce peak energy consumption
- ④ Dispatch energy resources and monitor peak shaving indicators

- ◆ 分析综合需求响应系统的典型应用场景中电、气、热及综合需求响应系统应用功能及服务架构，**设计面向综合能源服务的综合需求响应整体解决方案。**
- ◆ 以微功耗、高可靠性为目标，进行综合需求响应信息互动终端的设计，提出支持多类型通信协议接入的映射与交互技术，**研究综合需求响应信息互动终端的快速协调就地运行控制策略**，以及与系统主站及终端设备的数据处理技术。
- ◆ **设计综合能源服务模式**下综合需求响应软件应用功能，包括综合需求响应软件的信息标准化建模及数据预处理、数据融合与分析、软件灵活植入、需求侧资源可视化调度、模块灵活组态配置等。
- ◆ 研发响应的软硬件系统，开展基于电热气多能源互联的综合需求响应**应用验证**。

- ◆ **Analyze the application functions and service architecture of electricity, gas, heat and integrated demand response system in the typical application scenarios of integrated demand response system, and design the overall solution of integrated demand response for integrated energy services.**
- ◆ **Aiming at micro power consumption and high reliability, design the integrated demand response information interaction terminal, propose the mapping and interaction technology supporting multi type communication protocol access, and study the fast coordinated local operation control strategy** of the integrated demand response information interaction terminal and the data processing technology of the system master station and terminal equipment.
- ◆ **Design the application functions of integrated demand response software under integrated energy service mode**, including information standardization modeling and data preprocessing, data fusion and analysis, flexible software implantation, visual scheduling of demand side resources, flexible module configuration, etc.
- ◆ **Develop responsive software and hardware systems, and carry out comprehensive demand response application verification** based on electrothermal gas multi energy interconnection.

Key technologies 5: **situational awareness, non-intrusive measurement, and Internet of Things technology**

新型电力系统关键技术之五：态势感知、非侵入式测量与物联网技术

Deep perception of the operating status of massive energy-consuming devices, such as electricity, heat, gas, water, and transportation, will enable early prediction of system operational conditions.

支撑新型电力系统灵活运行的重要基础是对电、热、气、水、交通等海量用能装置运行状态的深度感知，以提前预判系统的运行态势。

- ◆ **Energy IoT technology** enables digitalization and intelligence at energy nodes, interacting and coordinating control through cloud-based IoT buses.
- ◆ **Non-intrusive measurement technology** analyzes the operating status of users' energy-consuming devices through high-precision node measurements. Edge computing processing chips embed data processing, algorithms, and communication interfaces.
- ◆ **5G technologies** such as high speed, reliability, and low latency are being applied in new power systems.
- ◆ **Beidou's location**, short messages, and precise timing services are applied in power system planning, operation, safety supervision, equipment management, and other fields.

Key technologies-6: The operational and trading mechanism of the market

新型电力系统关键技术之六：市场的运行交易机制

- ◆ Based on the different characteristics of regional networks and energy trading objects, implement peer-to-peer (P2P) trading for grid-connected users.
- ◆ Formulate the market's price formation mechanism, congestion management mechanism, and deviation energy balancing mechanism.
- ◆ Driven by market prices, form "autonomous trading and active regulation" to maximize and economic benefits.
- ◆ 根据区域网络以及交易能源对象的不同特点，实现入网用户的对等交易(P2P交易)。
- ◆ 制定市场的价格形成机制、阻塞管理机制和偏差能量平衡机制。
- ◆ 通过市场价格驱动，形成“自主交易、主动调节”，实现社会效益与经济效益的最大化。

Blockchain technology supports peer-to-peer transactions for massive users, enabling trustworthy transactions

- ◆ Through distributed general ledger data, stakeholders can interact peer-to-peer to achieve enforced trust
- ◆ Power transactions are automatically executed through smart contracts, enabling automatic response to demand fluctuations
- ◆ Achieve direct securitization of power plant revenues, enabling power plant crowdfunding through an equity platform

区块链
交易

区块链技术 支撑海量用户的点对点交易，实现可信交易

- ◆通过分布式总账数据，相关方可以点对点互动，实现强制信任
- ◆通过智能合约自动执行电力交易，需求波动能够得到自动响应
- ◆实现电站收益直接证券化，基于股权平台可实现电站众筹



THANKS

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